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```
import pandas as pd
import numpy as np
s = pd.Series([1, 3, 5, np.nan, 6, 8])
s
dates = pd.date_range("20130101", periods=6)
dates
```

```
DatetimeIndex(['2013-01-01', '2013-01-02', '2013-01-03', '2013-01-04',
               '2013-01-05', '2013-01-06'],
              dtype='datetime64[ns]', freq='D')
```

```
import pandas as pd
import numpy as np
df = pd.DataFrame(np.random.randn(6, 4), index=dates, columns=list("ABCD"))

df
```

	A	B	C	D
2013-01-01	2.438309	-0.533176	1.889585	0.768839
2013-01-02	-1.278178	1.980355	0.087232	0.257254
2013-01-03	0.793269	0.813440	0.397283	-0.549010
2013-01-04	0.456985	-1.824791	0.705465	0.239551
2013-01-05	-0.909268	1.096913	-0.731259	-0.887379
2013-01-06	0.580815	1.291639	-0.423542	1.049427

```
import pandas as pd
import numpy as np
df2 = pd.DataFrame(
    {
        "A": 1.0,
        "B": pd.Timestamp("20130102"),
        "C": pd.Series(1, index=list(range(4)), dtype="float32"),
        "D": np.array([3] * 4, dtype="int32"),
        "E": pd.Categorical(["test", "train", "test", "train"]),
        "F": "foo",
    }
)


df2
df2.dtypes
```

	0
A	float64
B	datetime64[s]
C	float32
D	int32
E	category
F	object

dtype: object

```
df.head()
```

	A	B	C	D
2013-01-01	2.438309	-0.533176	1.889585	0.768839
2013-01-02	-1.278178	1.980355	0.087232	0.257254
2013-01-03	0.793269	0.813440	0.397283	-0.549010
2013-01-04	0.456985	-1.824791	0.705465	0.239551
2013-01-05	-0.909268	1.096913	-0.731259	-0.887379

```
df.tail(2)
```

	A	B	C	D
2013-01-05	-0.909268	1.096913	-0.731259	-0.887379
2013-01-06	0.580815	1.291639	-0.423542	1.049427

```
df.index
```

```
DatetimeIndex(['2013-01-01', '2013-01-02', '2013-01-03', '2013-01-04',
               '2013-01-05', '2013-01-06'],
              dtype='datetime64[ns]', freq='D')
```

```
df.columns
```

```
Index(['A', 'B', 'C', 'D'], dtype='object')
```

```
df.to_numpy()
```

```
array([[ 2.43830855, -0.53317583,  1.88958463,  0.76883928],
       [-1.27817804,  1.98035469,  0.0872317 ,  0.25725371],
       [ 0.79326943,  0.81344022,  0.39728256, -0.54900962],
       [ 0.45698507, -1.82479116,  0.70546508,  0.23955136],
       [-0.90926817,  1.09691342, -0.73125909, -0.88737896],
       [ 0.58081482,  1.29163948, -0.42354221,  1.04942741]])
```

```
df2.dtypes
```

	0
A	float64
B	datetime64[s]
C	float32
D	int32
E	category
F	object

dtype: object

```
df2.to_numpy()
```

```
array([[1.0, Timestamp('2013-01-02 00:00:00'), 1.0, 3, 'test', 'foo'],
       [1.0, Timestamp('2013-01-02 00:00:00'), 1.0, 3, 'train', 'foo'],
       [1.0, Timestamp('2013-01-02 00:00:00'), 1.0, 3, 'test', 'foo'],
       [1.0, Timestamp('2013-01-02 00:00:00'), 1.0, 3, 'train', 'foo']],
      dtype=object)
```

displays quick Summary

```
df.describe()
```

	A	B	C	D
count	6.000000	6.000000	6.000000	6.000000
mean	0.346989	0.470730	0.320794	0.146447
std	1.331083	1.395683	0.930627	0.745137
min	-1.278178	-1.824791	-0.731259	-0.887379
25%	-0.567705	-0.196522	-0.295849	-0.351869
50%	0.518900	0.955177	0.242257	0.248403
75%	0.740156	1.242958	0.628419	0.640943
max	2.438309	1.980355	1.889585	1.049427

Transposing your data:

```
df.T
```

	2013-01-01	2013-01-02	2013-01-03	2013-01-04	2013-01-05	2013-01-06
A	2.438309	-1.278178	0.793269	0.456985	-0.909268	0.580815
B	-0.533176	1.980355	0.813440	-1.824791	1.096913	1.291639
C	1.889585	0.087232	0.397283	0.705465	-0.731259	-0.423542
D	0.768839	0.257254	-0.549010	0.239551	-0.887379	1.049427

DataFrame.sort_index() sorts by an axis:

```
df.sort_index(axis=1, ascending=False)
```

	D	C	B	A
2013-01-01	0.768839	1.889585	-0.533176	2.438309
2013-01-02	0.257254	0.087232	1.980355	-1.278178
2013-01-03	-0.549010	0.397283	0.813440	0.793269
2013-01-04	0.239551	0.705465	-1.824791	0.456985
2013-01-05	-0.887379	-0.731259	1.096913	-0.909268
2013-01-06	1.049427	-0.423542	1.291639	0.580815

DataFrame.sort_values() sorts by values:

```
df.sort_values(by="B")
```

	A	B	C	D
2013-01-04	0.456985	-1.824791	0.705465	0.239551
2013-01-01	2.438309	-0.533176	1.889585	0.768839
2013-01-03	0.793269	0.813440	0.397283	-0.549010
2013-01-05	-0.909268	1.096913	-0.731259	-0.887379
2013-01-06	0.580815	1.291639	-0.423542	1.049427
2013-01-02	-1.278178	1.980355	0.087232	0.257254

passing a single label selects a columns and yields a Series equivalent to df.A:

```
df["A"]
```

	A
2013-01-01	2.438309
2013-01-02	-1.278178
2013-01-03	0.793269
2013-01-04	0.456985
2013-01-05	-0.909268
2013-01-06	0.580815

dtype: float64

For a DataFrame, passing a slice : selects matching rows:

```
df[0:3]
```

	A	B	C	D
2013-01-01	2.438309	-0.533176	1.889585	0.768839
2013-01-02	-1.278178	1.980355	0.087232	0.257254
2013-01-03	0.793269	0.813440	0.397283	-0.549010

```
df["20130102":"20130104"]
```

	A	B	C	D
2013-01-02	-1.278178	1.980355	0.087232	0.257254
2013-01-03	0.793269	0.813440	0.397283	-0.549010
2013-01-04	0.456985	-1.824791	0.705465	0.239551

Selecting a row matching a label:

```
df.loc[dates[0]]
```

	2013-01-01
A	2.438309
B	-0.533176
C	1.889585
D	0.768839

dtype: float64

Selecting all rows (:) with a select column labels:

```
df.loc[:, ["A", "B"]]
```

	A	B
2013-01-01	2.438309	-0.533176
2013-01-02	-1.278178	1.980355
2013-01-03	0.793269	0.813440
2013-01-04	0.456985	-1.824791
2013-01-05	-0.909268	1.096913
2013-01-06	0.580815	1.291639

For label slicing, both endpoints are included:

```
df.loc["20130102":"20130104", ["A", "B"]]
```

	A	B
2013-01-02	-1.278178	1.980355
2013-01-03	0.793269	0.813440
2013-01-04	0.456985	-1.824791

```
df.loc[dates[0], "A"]
```

```
np.float64(2.438308551754381)
```

```
df.at[dates[0], "A"]
```

```
np.float64(2.438308551754381)
```

```
df.iloc[3]
```

	2013-01-04
A	0.456985
B	-1.824791
C	0.705465
D	0.239551

dtype: float64

```
df.iloc[3:5, 0:2]
```

	A	B
2013-01-04	0.456985	-1.824791
2013-01-05	-0.909268	1.096913

```
df.iloc[[1, 2, 4], [0, 2]]
```

	A	C
2013-01-02	-1.278178	0.087232
2013-01-03	0.793269	0.397283
2013-01-05	-0.909268	-0.731259

```
df.iloc[1:3, :]
```

	A	B	C	D
2013-01-02	-1.278178	1.980355	0.087232	0.257254
2013-01-03	0.793269	0.813440	0.397283	-0.549010

```
df.iloc[:, 1:3]
```

	B	C
2013-01-01	-0.533176	1.889585
2013-01-02	1.980355	0.087232
2013-01-03	0.813440	0.397283
2013-01-04	-1.824791	0.705465
2013-01-05	1.096913	-0.731259
2013-01-06	1.291639	-0.423542

```
df.iloc[1, 1]
```

```
np.float64(1.9803546903459956)
```

```
df.iat[1, 1]
```

```
np.float64(1.9803546903459956)
```

```
df[df["A"] > 0]
```

	A	B	C	D
2013-01-01	2.438309	-0.533176	1.889585	0.768839
2013-01-03	0.793269	0.813440	0.397283	-0.549010
2013-01-04	0.456985	-1.824791	0.705465	0.239551
2013-01-06	0.580815	1.291639	-0.423542	1.049427

```
df[df > 0]
```

	A	B	C	D
2013-01-01	2.438309	NaN	1.889585	0.768839
2013-01-02	NaN	1.980355	0.087232	0.257254
2013-01-03	0.793269	0.813440	0.397283	NaN
2013-01-04	0.456985	NaN	0.705465	0.239551
2013-01-05	NaN	1.096913	NaN	NaN
2013-01-06	0.580815	1.291639	NaN	1.049427

```
df2
```

	A	B	C	D	E
2013-01-01	2.438309	-0.533176	1.889585	0.768839	one
2013-01-02	-1.278178	1.980355	0.087232	0.257254	one
2013-01-03	0.793269	0.813440	0.397283	-0.549010	two
2013-01-04	0.456985	-1.824791	0.705465	0.239551	three
2013-01-05	-0.909268	1.096913	-0.731259	-0.887379	four
2013-01-06	0.580815	1.291639	-0.423542	1.049427	three

```
df2 = df.copy()
df2["E"] = ["one", "one", "two", "three", "four", "three"]
df2
```

	A	B	C	D	E
2013-01-01	2.438309	-0.533176	1.889585	0.768839	one
2013-01-02	-1.278178	1.980355	0.087232	0.257254	one
2013-01-03	0.793269	0.813440	0.397283	-0.549010	two
2013-01-04	0.456985	-1.824791	0.705465	0.239551	three
2013-01-05	-0.909268	1.096913	-0.731259	-0.887379	four
2013-01-06	0.580815	1.291639	-0.423542	1.049427	three

```
df2[df2["E"].isin(["two", "four"])]
```

	A	B	C	D	E
2013-01-03	0.793269	0.813440	0.397283	-0.549010	two
2013-01-05	-0.909268	1.096913	-0.731259	-0.887379	four